ASSIGNMENT 02

Supplementary Test Evidence

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Project: Sensor Monitoring System

Purpose: this addendum records additional results gathered after the original report was generated. It supplements the original 15-page report and does not replace or silently alter its contents.

Environment limitation: the memory, strace, perf, Valgrind, and jitter measurements described here were collected on the Ubuntu host using the host test program or the jitter test utility. They must not be interpreted as measurements performed on the QEMU target.

1. Memory footprint

The following measurements were recorded for the Ubuntu host process `test_host/sms_app_test`, using CSV test input `/tmp/sms_sensor_test.csv` during a 60-second run.

Elapsed time	VmSize (kB)	VmRSS (kB)
5 seconds	166636	2092
30 seconds	166636	2092
55 seconds	166636	2092

Observation: VmSize and VmRSS remained unchanged at the three sampled times in this run. These samples alone do not prove the absence of all memory leaks. Source: evidence/memory/memory_test_summary.txt.

2. Real-time jitter histogram

The histogram below was calculated from the existing CSV files. Values are classified by positive jitter in nanoseconds, with bins reported in microseconds (µs). Each mode contains 1,000 valid samples.

Jitter range	SCHED_OTHER	SCHED_FIFO
0–100 µs	970 (97.00%)	1000 (100.00%)
100–500 µs	28 (2.80%)	0 (0.00%)
500–1000 µs	2 (0.20%)	0 (0.00%)
>1000 µs	0 (0.00%)	0 (0.00%)
Total	1000	1000

In the recorded data, all SCHED_FIFO samples were below 100 µs, while 30 SCHED_OTHER samples were at least 100 µs. This describes these particular host measurements only and is not a real-time guarantee for the target system. Source: evidence/realtime/jitter_histogram.txt.

3. Additional SCHED_FIFO run

Metric	Recorded result
Environment	Ubuntu host
Command	sudo chrt -f 50 ./realtime/jitter_test FIFO /tmp/jitter_fifo_chrt.csv
Samples	1000
Period	1 ms
mlockall	success
Minimum jitter	7716 ns
Maximum jitter	100349 ns
Average jitter	33440.09 ns

The test completed and printed the FIFO policy and statistics. The result is stored in evidence/realtime/jitter_fifo_chrt.csv, evidence/realtime/jitter_fifo_chrt.log, and evidence/realtime/scheduler_test_summary.txt. This additional run is separate from the earlier jitter measurements; do not combine its statistics with the earlier dataset.

4. strace and perf evidence

The recorded host strace summary reports 116 system calls, 5 errors, and 0.009177 seconds total system-call time. The largest reported time shares were mq_timedreceive (30.38%), futex (17.67%), execve (10.68%), clock_nanosleep (9.47%), and clone3 (4.98%). These are host CSV-simulator results; errors should be interpreted in the context of that test setup. Source: evidence/strace_summary.txt.

The available perf evidence contains perf-stat measurements. A verified sampled perf report/hotspot analysis is not included in this addendum, so no hottest function is claimed. Source: evidence/perf_test_summary.txt and evidence/perf_stat_rerun_summary.txt.

5. Remaining evidence limitations

The available files do not establish a complete GDB transcript or a recorded before/after comparison of /proc/sms_stats. The GDB summary records that the main and four thread-function breakpoints were reached, and that the application exited normally, but it is not a full debugger transcript. Do not claim these missing artifacts were captured.

Log rotation has a recorded PASS result in QEMU. Other host-side measurements must remain clearly distinguished from QEMU target tests. Any additional demonstration or submission requirements should follow the instructor's final instructions.

Evidence files are retained in the project directory under `evidence/`. This addendum supplements the original report; it does not certify that every rubric item has been completed.